

NumPy Interview Questions and Answers

1. What is NumPy?

NumPy is a Python library used for numerical computing and array operations.

2. Why do we use NumPy?

We use NumPy for fast calculations, array operations, mathematical functions, and machine learning tasks.

3. Difference between Python List and NumPy Array?

Python lists are slower and use more memory, while NumPy arrays are faster and memory efficient.

4. How do you import NumPy?

```
import numpy as np
```

5. What is a NumPy array?

A NumPy array is a collection of similar data types stored in contiguous memory locations.

6. Difference between ndim, shape, and size?

ndim gives dimensions, shape gives rows/columns, and size gives total elements.

7. What is slicing in NumPy?

Slicing is used to access a portion of an array.

8. What is broadcasting in NumPy?

Broadcasting allows operations on arrays with different shapes.

9. What is vectorization?

Vectorization means performing operations without loops for faster execution.

10. What is reshape() in NumPy?

reshape() changes the dimensions of an array without changing the data.

11. Difference between reshape() and flatten()?

reshape() changes dimensions while flatten() converts the array into 1D.

12. Use of zeros() and ones()?

zeros() creates arrays of 0s and ones() creates arrays of 1s.

13. How do you find mean in NumPy?

Use np.mean(array).

14. How do you find maximum and minimum values?

Use np.max(array) and np.min(array).

15. Difference between copy() and view()?

copy() creates a separate array while view() shares original data.

16. What is random module in NumPy?

It is used to generate random numbers.

17. What is concatenate()?

concatenate() joins two arrays together.

18. What is the use of axis in NumPy?

axis defines the direction of operation.

19. Advantages of NumPy?

Fast performance, less memory usage, multidimensional arrays, and mathematical operations.

20. Where is NumPy used?

AI, Machine Learning, Data Science, Scientific Computing, and Deep Learning.